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A Three-Part Framework for Self-Regulated Personality Development across Adulthood

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Abstract: Recently, researchers interested in personality development have begun to acknowledge the roles of motivation and self-regulation for why traits change across adulthood. We propose three preconditions under which individuals may change their own levels of a personality trait through self-directed efforts. Firstly, individuals need to desire changing their trait-related behaviours either as an end in itself or in order to achieve other goals. Secondly, they need to consider behavioural changes feasible and be able to implement the desired changes. Thirdly, behavioural changes need to become habitual in order to constitute a stable trait. After elaborating on these three conditions, we review evidence attesting to the importance of motivation and self-regulation for trait development. We conclude with a discussion of the mutual interdependence of traits and goals, as well as the limits of self-regulated personality change. From our framework, we derive why personality changes across adulthood tend to be small to medium only, namely because they may require that all three preconditions for self-regulated personality change are fulfilled. We provide reasons for why people might not view change as desirable, feasible or fail to maintain it over time. Finally, we propose ideas for potential study designs to research self-regulated personality change.
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Key words: personality development; self-regulation; motivation; goals; expectancy; value

Browsing the self-help section in a book store is an interesting experience: An impressive number of guides with titles such as 'The 7 habits of highly effective people: Powerful lessons in personal change' (Covey, 2004) or 'Making a change for good: A guide to compassionate self-discipline' (Huber, 2007) reflects not only many people's desire to change or even transform their personalities but also bears the promise that self-change is possible. But how much is the development of our personalities really in our own hands? Can we get rid of unwanted characteristics and adopt traits that we consider more desirable? Can we really change for the better if we set our minds to it?

Personality theorists have become increasingly interested in the roles of motivation and self-regulation for personality (e.g. Denissen & Penke, 2008; Hoyle & Sherrill, 2006; Morf, 2006). Recently, researchers interested in personality *development* have begun to also incorporate motivational factors in their research designs (e.g. Bleidorn et al., 2010; Lüdtke, Trautwein, & Husemann, 2009; Roberts, Wood, & Caspi, 2008). We consider this paper to be part of an emerging line of theories (e.g. Denissen, Penke, & Wood, under review; Denissen, van Aken, Penke, & Wood, in press; Wood & Denissen, in press), which argue that a comprehensive model of personality development needs to take into account the active role of the individual, that is, processes of motivation and self-regulation, to explain personality stability and change.

In this paper, we propose a three-part framework that addresses how people actively shape their own personalities by keeping dispositional tendencies that they consider instrumental to their goals and overcoming dispositional tendencies that they consider undesirable and problematic.

A FUNCTIONAL PERSPECTIVE ON PERSONALITY DEVELOPMENT

In order to entertain the idea that motivation and self-regulation impact trait development, it is necessary to move away from a purely descriptive understanding of traits and take a functional perspective (e.g. Denissen & Penke, 2008; McCabe & Fleeson, 2012; Morf, 2006). We put forth that one way to understand personality is to look at the goals people try to achieve through their trait-related behaviours and at how self-regulatory efforts and habit formation processes might help to turn these goals into trait change. Before describing the conditions under which motivated and self-regulated personality development may occur, a definition of our main constructs is in order.

Self-regulation can be defined as occurring when people are modifying their own behaviour (including thoughts and feelings) according to goals or standards (Baumeister & Heatherton, 1996). People employ self-regulatory mechanisms when they experience discrepancies between a current

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(behavioural, cognitive and affective) state and valued internal, internalized or external standards (Higgins, 1987). Goals may serve as such standards for personality development.

Goals are cognitive representations of personally desired (or dreaded) end-states to be approached (or avoided) through action (Kruglanski, 1996). They vary in terms of their expectancy, and individuals can succeed or fail in attaining them. As goals are (unlike implicit motives such as the needs for achievement, power or affiliation) generally consciously accessible and can be expressed through language, they offer a useful level of analysis for the study of intentional personality development. Moreover, whereas many other motivational concepts describe personal preferences for outcome states in relatively broad and abstract terms (e.g. implicit motives, needs and ideal selves), the goal concept is silent with regard to its level of abstraction or specificity. Goals may be very concrete (e.g. 'I want to introduce myself to my new colleague tomorrow.'), very abstract ('I want people to like me.') or somewhere in between ('I want to get along with my colleagues at work.'; for example, Austin & Vancouver, 1996).

Goals may stem from intrinsic or extrinsic sources. People may feel the need to change because of external pressures, they may internalize external demands and feel an internal pressure to change while still not acting out of enjoyment or interest or they may be truly intrinsically motivated and view changing a trait (e.g. being more emotionally stable) as a desirable end in itself (Deci & Ryan, 1985). Irrespective of the source of motivation, the term self-regulation refers to the processes that are in action after a person has decided that change is both desirable or necessary and feasible. As we will elaborate, self-regulation requires the ability to identify goal-relevant situations, the investment of self-control or willpower to replace unwanted behaviour with desired behaviours, the monitoring of one's goal progress and the adjustment of effort.

THREE PRECONDITIONS FOR SELF-REGULATED PERSONALITY CHANGE

In our three-part framework, we propose that self-regulated personality development may occur under three conditions (Figure 1). Firstly, a person must consider new trait-related behaviours desirable as ends in themselves, or as instrumental or necessary means to advancing a superordinate goal (or goals). Secondly, a person must consider performing the new trait-related behaviour feasible and *actually* be capable of doing so. These two conditions can be roughly expressed by the terms *value* and *expectancy*, which together determine a person's commitment to and likelihood of success at performing/omitting behaviours in classic models of motivation and planned action (e.g. Atkinson, 1957; Fishbein & Ajzen, 1975; Heckhausen, 1977; Vroom, 1964). Thirdly, only if a person frequently enacts the respective trait-related behaviours, and thereby turns the self-regulated changes into habits should these forces be regarded as resulting in a stable shift in a person's personality. We will now discuss the three conditions that manifest the core of our framework in more detail.

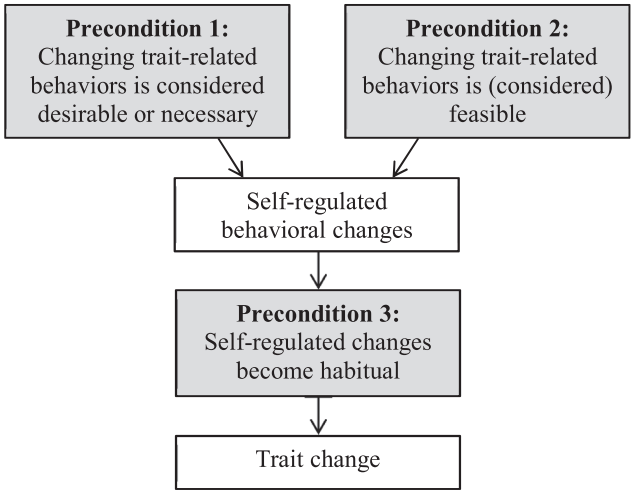

Figure 1. Three-part framework for self-regulated personality development. Note that Preconditions 1 and 2 are necessary for the self-regulated initiation of behavioural change, whereas Precondition 3 is necessary for behavioural change to be maintained and turn into a stable trait.

Precondition 1: Changing trait-related behaviours is considered desirable or necessary

What people consider a desirable trait is the product of many influences, such as biological dispositions, social norms, idiosyncratic histories of conditioning and environmental demands all of which may change over the lifespan (e.g. Eagly & Chaiken, 1993; Merton & Kitt, 1952; Skinner, 1974). Independent of *why* people consider certain end-states as desirable, if personality development has an intentional component, it should follow the direction that individuals consider desirable. People may value certain trait-related behaviours either as ends in themselves (e.g. being emotionally stable) or as instrumental means for the pursuit of other higher level goals. This pertains to both approach and avoidance goals. Regarding the former, a person might, for example, consider acting friendly and cooperatively (behaviours associated with the trait of agreeableness) as instrumental means for attaining the superordinate goal to get along well with colleagues at work, which in turn, might help to have a harmonious and productive work situation (McCabe & Fleeson, 2012). Regarding the latter, a person might consider a trait as an instrumental means for avoiding undesired states. For example, a person might display behaviours associated with the trait of conscientiousness in order avoid the stress and potential failure that comes from procrastinating dreaded but necessary work tasks until the last minute.

Independent of whether trait-related behaviours are represented as ends in themselves or as means to other goals, how broad and abstract the respective goal is should have implications for how broadly behaviour is affected across situations and time. For example, abstract goals such as achieving 'emotional stability' should affect most situations that are emotionally relevant. Similarly, an abstract goal such as 'getting along with and being liked by others' should result in behaviours that a person considers instrumental to achieve this aim, for example, being courteous, cooperative and friendly, across most situations that involve other people.

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More specific goals, such as 'get along with one's boss', however may affect actions in specific situations only, here at the workplace. Likewise, people might consider changing only facets of a trait (e.g. being more punctual) necessary.

Precondition 2: Changing trait-related behaviours is (considered) feasible

The second condition to achieve intentional personality change is that it needs to be considered feasible. In addition to the desirability of trait-related behaviours (i.e. their positive and negative consequences), their subjective expectancy or feasibility also has an impact on whether the individual will try to perform them or not. It has been shown that people hold different beliefs about whether personality is fixed or malleable (e.g. Dweck, 2008). Most adults believe that they can actively change their personalities (Chiu, Hong, & Dweck, 1997; Noftle, 2013). Other studies have shown that interventions that target people's beliefs about the malleability of personality traits promote change in these traits. For example, an intervention that increased children's beliefs that aggression is a malleable rather than a fixed trait successfully reduced children's aggressive behaviour and conduct problems for an extended period when compared with a control group without treatment (Yeager, Trzesniewski, & Dweck, 2013).

People's subjective beliefs about the feasibility of enacting particular behaviours may be more or less accurate reflections of their *actual capacity* to override habitual responses and substitute them with desired responses. This actual capacity will ultimately determine whether they can implement desired changes. Based on general models of self-regulation (e.g. Carver & Scheier, 1990), we propose a three potential ingredients of this capacity. Generally, if individuals are committed to a goal, they perceive their environment in terms of affordances (e.g. Bruner & Goodman, 1947; Vogt, De Houwer, Moors, Van Damme, & Crombez, 2010). The first ingredient is therefore the ability to correctly identify goal-relevant situations in which habitual responses can be replaced with others. We are not aware of research that directly addresses individual differences in this ability, although it might be possible to draw some lessons from the literature on individual differences in entrepreneurs' handling of opportunities and propensity to take action. In this literature, alertness and prior knowledge have been proposed as necessary preconditions for the successful recognition of action opportunities. People who are alert to search the environment for opportunities and who have encountered and successfully seized action opportunities in the past should be more likely to successfully identify affordances in the future (Ardichvili, Cardozo, & Ray, 2003). People also need to have the necessary means at their command, that is, know responses that can replace old undesired responses (Arkowitz, 2002; see our discussion of the limits of change in the succeeding text). A second ingredient is willpower or self-control to override a habitual but unwanted response and substitute it with a desired response (Baumeister, Vohs, & Tice, 2007). Individual differences in self-control have been shown to predict how successfully people master their tasks and live their lives (e.g. Mischel, Shoda, & Rodriguez, 1989; Moffitt et al., 2011; Tangney, Baumeister, & Boone, 2004).

Monitoring goal progress in order to make adjustments to one's new behavioural responses (e.g. to redouble one's effort) constitutes a third ingredient of successful self-regulation. The most prominent process models of self-regulation (e.g. Carver & Scheier, 1990; Miller, Galanter, & Pribram, 1960) operationalize goal-directed behaviour in terms of discrepancy-reducing processes between a current state and a referent state (e.g. an 'ideal' state). In these models, monitoring processes or entities make sure that any discrepancy between a current state and a referent standard is detected and that effort is expended until the current state matches the referent state. People vary in the degree to which they attend to how well their behaviour matches their standards (e.g. Duval & Wicklund, 1972; Schraw & Dennison, 1994; Snyder, 1974), which may be an interesting predictor of how successfully people can implement desired personality changes. People can report on their own propensity for self-monitoring and these self-reports have been shown to predict actual performance (Caldwell & O'Reilly, 1982; Schraw, 1994).

Precondition 3: Self-regulated changes become habitual

Although preconditions 1 and 2 determine whether a person will set the goal to change in the first place, precondition 3 concerns the processes by which an initial behavioural change turns into a trait. We propose that processes of habit formation help a person to maintain the desired behaviour changes across time and, as a consequence, translate them into relatively stable and measurable trait changes. All skills (e.g. driving and typing) require effortful and intentional action before becoming efficient and reflexive through repetition and experience (Newell & Rosenbloom, 1981). The behavioural tendencies that identify personality traits like extraversion (e.g. tendencies to be talkative, sociable and assertive) are no exception and become efficient and reflexive through repetition. Trait-identifying behaviours can become automatized if a specific context has been repeatedly paired with goal-directed behaviour. After the association has been built, the context may later on trigger goal-directed action without the need for conscious, intentional intervention (e.g. Aarts & Dijksterhuis, 2000; Bargh & Gollwitzer, 1994; Bargh, Gollwitzer, Lee-Chai, Barndollar, & Troetschel, 2001). For example, after having received negative feedback from her professors, a student might want to overcome her lack of conscientiousness. In order to improve her grades and study performance, she makes a detailed plan that covers all her projects and tasks and assigns them different priority levels. In the beginning, she might have to force herself to consult and update her planned schedule every morning but on the long run doing so may become a habit. Through frequent context–behaviour pairings, the once effortful pursuit of the goal to be more organized and reliable has become automatized, unconscious and effortless (e.g. Danner, Aarts, & de Vries, 2008; Lally, Van Jaarsveld, Potts, & Wardle, 2010; Wood & Neal, 2007).

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EVIDENCE FOR SELF-REGULATED PERSONALITY CHANGE

We argue that normative personality changes across adulthood, such as the previously reported increases in conscientiousness, agreeableness, social dominance and emotional stability (Roberts, Walton, & Viechtbauer, 2006), are partly consequences of the goals people pursue across the lifespan, as well as age-normative changes in these goals. If personality change is indeed affected by motivational and self-regulatory processes, intra-individual trajectories of personality change should be influenced by the traits and end-states people consider desirable and attainable across adulthood. We now review three types of evidence that support the idea that personal goals affect trait development: (i) evidence suggesting that personality development might be mediated by shifts in trait desirabilities; (ii) evidence suggesting that the traits that show mean level increases across adulthood might be functional for the achievement of goals adults typically pursue across the lifespan and (iii) evidence based on longitudinal data directly linking goal development to trait development.

Evidence that personality development is associated with trait desirability

There are various indications that the same traits that show mean level increases with age also become increasingly valued with age. Specifically, meta-analytic results suggest that individuals tend to show age-related mean level increases in emotional stability, agreeableness and conscientiousness (Roberts et al., 2006). Assessing the desirability of traits, Wortman and Wood (2013) found that older adults describe agreeableness-related and conscientiousness-related traits such as being *dependable, competent, courteous* and *practical* as more desirable than young adults, and that these differences likely mediate some of the age-related increases in trait levels. Similarly, as people get older, they are increasingly expected to act in an agreeable, conscientious and stable manner (Chan et al., 2012; Wood & Roberts, 2006), and acting in a disagreeable and unconscientious manner becomes increasingly associated with feeling 'abnormal' in old age (Wood, Gosling, & Potter, 2007). Communal characteristics also appear to become increasingly attractive to potential mates in adulthood (Brumbaugh & Wood, 2013; Wood & Brumbaugh, 2009). Accordingly, the increase of communal behaviours in old age may partly be attributed to their increased desirability resulting from a change of norms. Moreover, Lehmann, Denissen, Allemand, and Penke (2013) found age differences in a measure designed to assess motivational content related to the Big Five (e.g. for agreeableness: 'I would rather share something than to compete with other people for having it all myself') to show similar age trajectories as more traditional Big Five measures, further indicating that age-related changes may be driven, in part, by changes in the value of outcomes that follow the performance of trait-related behaviour.

Although these results suggest that personality change may be driven by trait desirabilities, they cannot refute the possibility that older people may value how their traits have changed without having actively caused their change. Older adults may have realized that becoming more dependable or courteous with age has helped them to more successfully attain their goals. They may therefore appreciate dependability and courtesy more than younger adults who have not yet experienced how useful high levels of these traits are. The change itself, however, may have been caused by maturation processes that are not regulated by the self, but are, for example, biologically driven. Only longitudinal studies that measure and experiments that manipulate trait desirabilities would provide a good tests of whether individual trait desirabilities cause individual differences in trajectories of trait change (see 'Implications for future research' in the succeeding text).

Evidence that the traits that show mean level increases across adulthood are functional for the achievement of age-graded goals

Another possible source of evidence that personality change occurs in the service of goal achievement lies in the question of whether the observed increases in conscientiousness, emotional stability, agreeableness and social dominance are indeed functional for the achievement of the goals adults pursue across the lifespan. The goals of many younger-aged and middle-aged adults typically involve their occupations, as well as to the formation of romantic relationships and families (Nurmi, Pulliainen, & Salmela-Aro, 1992). With regard to occupational goals, higher levels of conscientiousness have indeed been found to be related to higher job performance levels across many occupations (e.g. Barrick & Mount, 1991; Hurtz & Donovan, 2000; Judge, Higgins, Thoresen, & Barrick, 1999). Similarly, higher levels of extraversion seem to foster job performance in professions involving frequent social interactions and openness and extraversion seems to be related related to higher training proficiency across professions (Barrick & Mount, 1991). With regard to relationships and family life, conscientiousness, emotional stability and agreeableness have frequently been reported to be positively related to relationship satisfaction (e.g. Caughlin, Huston, & Houts, 2000; Robins, Caspi, & Moffitt, 2000; but see Noftle & Shaver, 2006), the promotion of more positive parenting styles (for a review, see Prinzie, Stams, Dekovic, Reijntnes, & Belsky, 2009), better parent-child relationships (Kochanska, Friesenborg, Lange, & Martel, 2004), better-adjusted children (e.g. Kochanska, Aksan, Penney, & Boldt, 2007) and a superior mastering of parenting challenges (Hutteman et al., 2014). Accordingly, the observed changes in these traits can be deemed functional for typical goals of younger-aged and middle-aged adults.

Finally, in old age, people tend to increase in their striving for 'intrinsically rewarding' goals that concern self-acceptance, emotional intimacy or community contribution (Penningroth & Scott, 2012; Sheldon & Kasser, 2001)—a pattern that thematically relates to the fact that, with age, people become more confident, warm, responsible and calm (Roberts & Wood, 2006). In addition, coping with health declines become an important new goal in later life. Indeed, some of the traits that are higher in older adults than in

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younger adults, namely conscientiousness and emotional stability, have positive effects on health-relevant behaviours (e.g. Booth-Kewley & Vickers, 1994; Hill & Roberts, 2011), suggesting a functional value of trait changes for goal pursuit across the lifespan.

Evidence based on longitudinal data directly linking goal development to trait development

Only a small number of studies have looked directly at the effects of goals on trait development. Bleidorn et al. (2010) showed in a sample of middle-aged twins that the importance of agency and communion-oriented life goals (e.g. 'have prestigious positions' and 'be friends with many people') predicted personality changes in conscientiousness, openness and extraversion over a period of 5 years. The finding supports the idea that people adapt their personality traits to advance the pursuit of their valued goals.

In contrast, Lüdtke et al. (2009) reported almost no effects of students' major life goals in eight domains (e.g. relationships, health and personal growth) on personality development. There were prospective effects of prior personality traits on later life goal importance. For example, agreeableness predicted the importance of relationship goals, whereas openness predicted the importance of personal growth. However, almost no effects of goals at T1 on personality traits at T2 were found. However, as Lüdtke et al. (2009) noted, their study covered a period of only 2 years and both goals and traits changed little within this short period, which might have made it hard to detect their codevelopment.

Finally, Roberts, O'Donnell, and Robins (2004) assessed college students' life goals and personalities multiple times over a 4-year period and found changes in life goals to be associated with personality changes. More specifically, if students increased their endorsement of economic and relationship goals over time, their agreeableness and conscientiousness increased as well. Because these associations were at the level of correlated change, it is difficult to establish whether goal changes impacted trait changes or *vice versa*.

DISCUSSION

In the previous sections, we have specified the preconditions under which self-regulated personality change may occur. However, we have left a few questions unanswered. In the following, we discuss what we call the 'chicken-egg-problem', that is, the question of whether goals are more likely to influence traits or *vice versa*, the definition of how and when we can tell that a personality trait has truly changed, as well as the reasons that people may not desire or may fail to change their personalities. We furthermore propose designs for research studies that would be suited to test the empirical value of our framework.

The chicken-egg problem: Do goals influence traits or do traits influence goals?

Although throughout this paper we have argued that goals and their development can drive intentional trait development, the other causal direction—traits driving goals—certainly is likely to occur as well (e.g. Bleidorn et al., 2010; Kandler, Zimmermann, & McAdams, 2014; Lüdtke et al., 2009; Roberts et al., 2004). In many situations, people may commit to goals that are in line with their personalities, or de-invest from goals that are not. This overlap might occur mainly because people consider trait-consistent goals to be more desirable as well as more feasible (see our discussion of the limits of change in the succeeding text). For this reason, an introverted man is probably less likely than an extraverted man to commit to meet new friends by attending social gatherings. By committing goals that are already within their dispositional behaviour spectrum, and disinvesting from goals that are not, people might consolidate or 'deepen' their prevalent traits (Roberts et al., 2008). Situations in which goals cannot be achieved if a person remains within the behavioural spectrum of his trait should therefore be of particular interest for studying personality change.

What do we consider 'true' personality change?

One may ask, whether changing one's behaviour in the service of a goal reflects true personality change. If, for example, a man who thrives and recharges on time alone joins social groups and participates in their activities in order to find a mate, should we really consider him a new extravert? The definition of personality used in evaluating personality at any one time is the same as the definition of personality used in evaluating its change. If reliable personality measures that converge across situations and data sources (e.g. self-ratings, peer-ratings and behavioural indicators) detect a stable and consistent behavioural change, we can assume that personality has truly changed. Of course, the man and his peers may discount the new extraverted behaviour in their personality ratings, thinking that it does not indicate his *actual* personality as the behaviour is situation-specific and motivated by an extrinsic goal ('I/He only go/goes out to find a partner but will stop doing so as soon as I/he can.'). The man might indeed not continue acting extraverted after having found a partner. He may, however, continue acting extraverted because he realized that social activities are fun, because it became functional for other goals (e.g. to keep the new partner) or because it turned into a new habit. In these cases, he and his peers, including his new acquaintances who never considered him an introvert, will probably start to view him as an outgoing person, allowing personality surveys to pick up on the behavioural change. Studies addressing personality change should therefore incorporate multiple measurement occasions at long enough time intervals for peer raters to adjust to the target person's personality change, and to offer better evidence that personality changes have been sustained over extended periods.

The limits of self-regulated personality development: Why don't people change more?

In light of our notion that people can intentionally change themselves for the better, it may be surprising that people's personalities, in fact, change relatively little across adulthood.

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For example, rank order is preserved over long periods, and the observed mean level increases in agreeableness, conscientiousness, social dominance and emotional stability tend to be small to medium (Roberts & DelVecchio, 2000; Terracciano, Costa, & McCrae, 2006). The fact that people do not change more raises the question about the limits of self-regulated personality change. In our view, the fact that all three preconditions for self-regulated personality change presented in the previous text have to be fulfilled is likely the most important reason why personality change tends to be limited. Being motivated does not suffice if a person lacks the capacity to change, having the capacity to change will not result in change if no change is desired and a person who would like to change and has the capacity to change is not going to change without the confidence that they can. And having both the desire and ability to change may still not result in enduring trait change if a person is not afforded sufficient opportunities to practise the trait-related behaviours and thus turn them into habits. In addition, the following reasons may reduce the desirability and feasibility of personality change, as well as the feasibility of maintaining personality change across time (see also Wood & Denissen, in press).

Firstly, data suggest that most people view their current trait levels as relatively desirable (Wortman & Wood, 2013). Interestingly, personality stability may be increased by the fact that individuals make errors in forecasting the desirability of certain outcomes. A recent study has argued that such errors keep introverted people from acting more extraverted (Zelenski et al., 2013). Specifically, introverts underestimated the enjoyment they would derive from acting extraverted, despite indications that introverts receive just as much enjoyment from acting extraverted as do extraverts (Fleeson, Malanos, & Achille, 2002).

Secondly, in order to maintain positive self-views, people often attribute negative experiences and failures to external causes (e.g. Bradley, 1978; Miller, 1976), thereby keeping themselves from seeing potential benefits of self-change. People may, for example, attribute frequent experiences of social conflict to their social interaction partners rather than their own lack of agreeableness and therefore see no reason to change. Similarly, people with narcissistic personalities often fail to acknowledge how much their own behaviour contributes to their interpersonal problems and see other people or external circumstances as responsible for their dissatisfaction. If they seek therapy, it is important and often difficult to convince them that their dissatisfaction stems from their own maladaptive behaviours (Sachse, 2001).

Thirdly, people may not desire to and even be resistant to change even maladaptive behaviours because of their desire for coherence and self-consistency (Swann, Rentfrow, & Guinn, 2003). If they continue to act the way they always did, their behaviour has familiar and therefore predictable consequences. They will not be threatened by unexpected reactions of their environments (Arkowitz, 2002). As a consequence, their belief systems, the lenses through which they see reality, can be maintained, providing them with a sense of control and coherence (Swann, Stein-Seroussi, & Giesler, 1992). People may also be fearful of not being able to deal with the consequences of changing their behaviour. Although they may want to change, they may at the same time fear that change may bring with it new demands and expectations. A shy person might, for example, seek therapy in order to learn how to approach other people. At the same time, he might be fearful of being rejected or the inability to cope with interpersonal intimacy. Such motivational conflicts may also keep people from trying to change their behaviour (Arkowitz, 1989, 2002).

Fourthly, acting in line with one's dispositions might be easy, whereas overcoming one's dispositions is effortful (Gallagher, Fleeson, & Hoyle, 2011) and may therefore be aversive. A hedonic preference for trait-consistent behaviour might also be due to physiological parameters. For example, even in situations in which they realize that more extraverted behaviours would be useful, introverts' physiological configuration might render acting extraverted unpleasant and keep them from being more outgoing (Denissen & Penke, 2008; Lucas & Diener, 2001). Overall, a trait's desirability should not only be affected by a person's expectations about its extrinsic instrumentality for personal goals but also by a person's expectations about the 'intrinsic value' of the trait, the immediate pleasantness of performing trait-related behaviours. Moreover, maintaining trait-inconsistent behaviour that is perceived as highly aversive requires self-control that, on the long run, might be depleted (Baumeister et al., 2007), leading to lower subjective and objective feasibility of trait change. We address this possibility in more detail in the succeeding text. Fifthly, change may actually be unnecessary if people have been successful in selecting environments to which their personalities are naturally suited. According to Roberts et al. (2008), people's personalities will cause them to occupy specific environmental niches that fit their traits. In such corresponsive environments, people should feel that they fit in well. For example, open individuals are more likely to choose and then thrive in occupations that require high levels of creativity (e.g. being architects), whereas individuals low in openness are more likely to choose and thrive in occupations that favour routines (e.g. being government tax and excise officials; for empirical evidence, see Denissen, Ulferts, Lüdtke, Gerstorf, & Muck, under review). To the degree that these niches have stable affordances and requirements, they will stabilize the traits that led people to seek out the respective niches in the first place. Only changes in the affordances and requirements of these environments will make change appear desirable or necessary.

Turning to the second and third preconditions, we outlined as necessary for personality change (Figure 1); people may not consider personality change feasible or not have the resources and abilities to make it happen and to maintain it across time. As noted earlier, overcoming dispositions is effortful and costly. For example, it has been proposed that behaving in ways that are inconsistent with one's dispositions depletes self-regulatory resources, causing people to tend to gravitate back to their original dispositions (Baumeister et al., 2007; Gallagher et al., 2011). To the degree that individuals do not have the necessary self-regulatory resources such as energy, attention, time or skills that need to be invested into behavioural change, they may not be able to change. For example, a fully employed parent of two children

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who is also taking care of their aged parents may be too tied up in daily responsibilities even to consider self-improvement. People may also lack role models that show alternatives to their own maladaptive behavioural patterns. For example, a child growing up in an aggressive household may never get to see disagreements being resolved peacefully (see also Roberts et al., 2008).

Implications for future research

As yet, few studies have directly looked at the longitudinal codevelopment of goals and traits. Future research should test whether personality change can be explained within an expectancy × value framework. More specifically, studies would need to test whether individual differences in the subjective desirability of traits, as well as individual differences in the perceived feasibility of personality change predict individual trajectories of personality change. If our claims are valid, people who consider trait change more desirable and, at the same time, more feasible than others should change more in the direction they consider desirable and feasible than other people.

An interesting question is furthermore whether broadly shared similarities in people's personality trajectories can be explained by socially consensual trait desirabilities (e.g. cultural norms) or age-graded developments in self-regulatory ability (e.g. age-related increases in executive functioning). In other words, goals and self-regulatory abilities should, to the degree that they are normative, predict normative patterns of change. Goals and self-regulatory abilities that are against the norm should explain non-normative patterns of change, that is, an individual's deviation from the average trajectory of change. For example, individuals who have hitherto been following a typical ('normative') personality trajectories but who face sudden and idiosyncratic shifts in goals (e.g. upon entering a deviant social group) or ability (e.g. following brain injury) might show non-normative patterns of change. Likewise, individuals may be motivated to approach the norm but therefore have to change in a non-normative direction (e.g. if an adolescent becomes more conscientious while all of his peers become less conscientious).

What would an ideal research study testing our expectancy value approach to self-regulated personality development look like? Clearly, the study should employ a longitudinal design with the repeated measurement of personality traits (e.g. Bleidorn et al., 2010; Lüdtke et al., 2009; Roberts et al., 2004). To assess the proposed predictors of interindividual variation in self-regulated trait change, an 'ideothetic' approach should combine the idiographic open-format assessment of personal goals with the more standardized nomothethic assessment of the desirability and feasibility of trait change (e.g. Cox & Klinger, 2004; Emmons, 1986). Firstly, participants should indicate the changes they aspire to achieve (or avoid) within the time frame that is covered by the study (e.g. 'finding a partner' or 'getting promoted at work'). Participants should then rate (i) the desirability of trait change based on its functional value to achieve these goals (e.g. 'In order to find a partner, do you think it would be valuable to become more/less outgoing in the future?') and (ii) the degree to which they consider trait change feasible (e.g. 'How confident are you that you will be successful in being more outgoing in the future?'). Consistent with expectancy × value models (e.g. Atkinson, 1957; Fishbein & Ajzen, 1975; Heckhausen, 1977; Vroom, 1964), the product of value and feasibility should prospectively predict the amount of change on the respective trait.

Intervention studies that manipulate both the desirability and the subjective feasibility of trait change in control-group designs would offer informative additions to observational studies and a different and strong way to test for causality. Such studies could target subclinical populations who seek help to tackle a personal problem, for example, individuals who want to become more successful in their social relationships or at work. These individuals may receive external feedback about discrepancies between their actual and their desired personality profile to spur their desire to change specific traits that, in their own view, are relevant to their goals (e.g. becoming more assertive in social relationships or becoming more conscientiousness at work). Moreover, targeting implicit beliefs about the malleability of these traits should affect whether people change or not (see Yeager et al., 2013). In a 2 (discrepancy feedback vs no feedback) × 2 (malleability intervention vs no intervention) between-subjects design personality change should be most pronounced in the group that receives both discrepancy feedback relating problems to specific traits and participates in an intervention that highlights the malleability of behaviour and personality. A waiting control group may be offered the same treatment with some delay to distinguish the effects of time progression from actual treatment effects.

One area of research that has already looked at the effect of goals on self-directed change through interventions is clinical and psychotherapy research. If people decide that they need to change and seek out psychotherapy to receive help in doing so, the resulting change can be seen as self-regulated. As has been shown, psychotherapy is often effective in targeting specific problematic behaviours and can also lead to sustained personality changes on the trait level. For example, as reported in a meta-analysis by Jorm (1989), therapy can increase a person's level of emotional stability. Other researchers have looked at how goal orientation and the specificity of change goals and motives impact how much patients can benefit from clinical interventions (e.g. Michalak, Klappheck, & Kosfelder, 2004). Clinical psychologists have also developed techniques to resolve the ambivalence that keeps people from making the changes they hope for and fear at the same time (motivational interviewing, e.g. Burke, Arkowitz, & Menchola, 2003). Teaming up with clinical psychologists seems fruitful to inspire and inform our research on the preconditions and processes of self-regulated personality development. Moreover, their knowledge about the factors that cause therapeutic efforts to fail (e.g. lack of resources or resistance to change, Arkowitz, 2002; Grawe & Grawe-Gerber, 1999) can help us to understand the limits of self-regulated personality development.

To tackle the question to what degree normative personality change has intentional components, research should

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compare how cross-cultural differences in trait desirabilities and cultural beliefs about the feasibility of personality change shape personality development. Self-regulated normative change should follow a trajectory that is normatively viewed as desirable within a culture (see Bleidorn et al., 2013). Moreover, it should occur only if individuals within a culture hold the belief that changing oneself is possible. Differences between cultures could indicate self-regulated processes of change.

To also assess how the actual capacity to overcome one's dispositions moderates the effects of the desirability and the subjective feasibility of change on personality development, predictors such as a person's trait self-control or their monitoring skills should also be included in both observation and intervention studies. Finally, both should incorporate multiple (at least more than two) measurement occasions to account for the potentially complex developmental interplay between traits and goals (Luhmann, Orth, Specht, Kandler, & Lucas, 2014) and detect if people can maintain behavioural changes over time.

Within our self-regulatory framework of personality development, other interesting questions may be addressed. For example, more distal predictors of the desirability of and the subjective feasibility of personality change could be examined: What causes a person to desire personality change in the first place? What influences whether people believe that personality is malleable or fixed? Then, if people set out to change, what is required for them to be successful on the long run? Which internal and external resources, such as certain personality characteristics other than self-control (e.g. openness and intelligence) or sources of social support, predict successful personality development? And what are typical trajectories of change? Do people change quickly in the beginning but then slowly stabilize? Or does self-regulated personality change follow a linear trajectory? While in this article, we cannot go into detail about these and other potentially interesting research questions, we believe that our framework may be useful for inspiring hypotheses and organizing future research on the causes and processes of self-regulated personality development.

Conclusion

We have proposed a framework that specifies the conditions and processes through which self-regulated personality change can take place. Assessing if people consider specific traits desirable as ends in themselves or as means to external goals, assessing or increasing the degree to which people believe that they can change these traits, and assessing and improving people's capacities to self-regulate their behaviour (e.g. self-control and monitoring skills) are fruitful research strategies to predict personality change and shed further light on how people may intentionally develop their personalities in the directions they desire. We believe that our perspective complements existing models of personality development that, so far, have emphasized biological and social/environmental influences on personality development (e.g. McCrae & Costa, 2008; Roberts & Jackson, 2008). Although we concur that biological and social/environmental influences are likely causes of personality development, our framework focuses on the *processes* that underlie trait change and considers people as proactive contributors to their own development (see also Denissen, Penke, et al., under review; Fleeson, 2012). Our framework may act as a theoretical basis for future research on the causes and underlying processes of personality development across adulthood.

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